CLASSIFICATION

UNCLASSIFIED

Windows Log Analysis

Splunk SIEM — Brute Force Detection & Investigation

SOC PROJECT | SPLUNK 10.2.1 | WINDOWS SERVER 2022 | MITRE ATT&CK;

Prepared by

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SECTION 01

Executive Summary

This report documents a hands-on Windows Log Analysis project conducted in a controlled SOC home lab environment. The objective was to simulate a real-world SOC analyst workflow — configuring Splunk SIEM to ingest Windows Security Event Logs, establishing a baseline of normal activity, simulating a brute force attack from Kali Linux, detecting the attack through log investigation, identifying the attacker via log correlation, and building a professional monitoring dashboard for continuous security visibility.

The investigation successfully detected 626 failed logon events (EventCode 4625) generated by the Hydra brute force tool, traced all 3,326 failed attempts to the attacker workstation 'kali', confirmed no successful unauthorized access occurred, and produced a four-panel Splunk dashboard aligned to real enterprise SOC operations.

3,467	626	3,326	0
Total Events Collected	Failed Logins Detected	Attack Attempts Traced	Accounts Compromised

SECTION 02

Lab Architecture

Component	System	IP Address	Role
Attacker	Kali Linux 2025.2	192.168.52.133	Brute force simulation via Hydra
Target	Windows Server 2022	192.168.52.139	Windows Event Log generation
SIEM	Splunk Enterprise 10.2.1	127.0.0.1:8000	Log collection, search & dashboard
Platform	VMware Workstation Pro 17	Host Machine	Virtualization environment

SECTION 03

SOC Investigation Workflow

01	→	02	→	03	→	04	→	05
Log Collection		Baseline Analysis		Attack Simulation		Detection & Triage		Attacker Identified

SECTION 04

Step 1 — Log Collection & Verification

Splunk Enterprise was configured to ingest Windows Security, System, and Application Event Logs from Windows Server 2022 in real time by editing the `inputs.conf` configuration file to add the `WinEventLog://Security` stanza. After a clean service restart, log ingestion was confirmed via a Splunk search returning **3,467 events** across multiple sources — the essential first step of any SOC investigation.

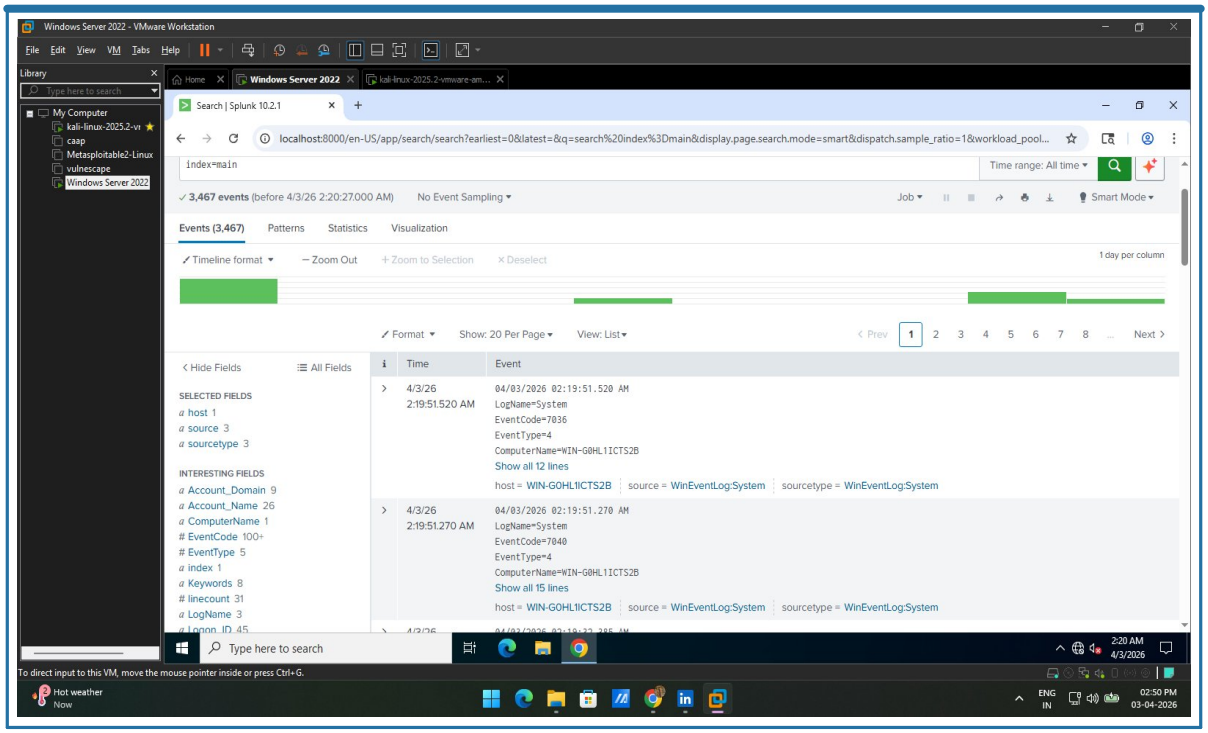

Figure 1 — Splunk Search confirming 3,467 Windows Event logs successfully ingested across Security, System and Application log sources in real time.

SECTION 05

Step 2 — Baseline Analysis (EventCode 4624)

Before investigating suspicious activity, a SOC analyst establishes what normal authentication looks like. A search for **EventCode 4624** (Successful Logon) returned **345 events** from the Windows Security log, revealing the expected accounts and login patterns across the system.

Account	Login Count	Type	Assessment
WIN-G0HL1ICTS2B$	263	Machine Account	**Normal — automated system auth**
SYSTEM	228	System Account	**Normal — internal OS operations**
Splunkd	9	Service Account	**Normal — SIEM service logins**

Administrator	4	Admin Account	**Normal — manual admin sessions**
LOCAL SERVICE	4	Service Account	**Normal — Windows service**

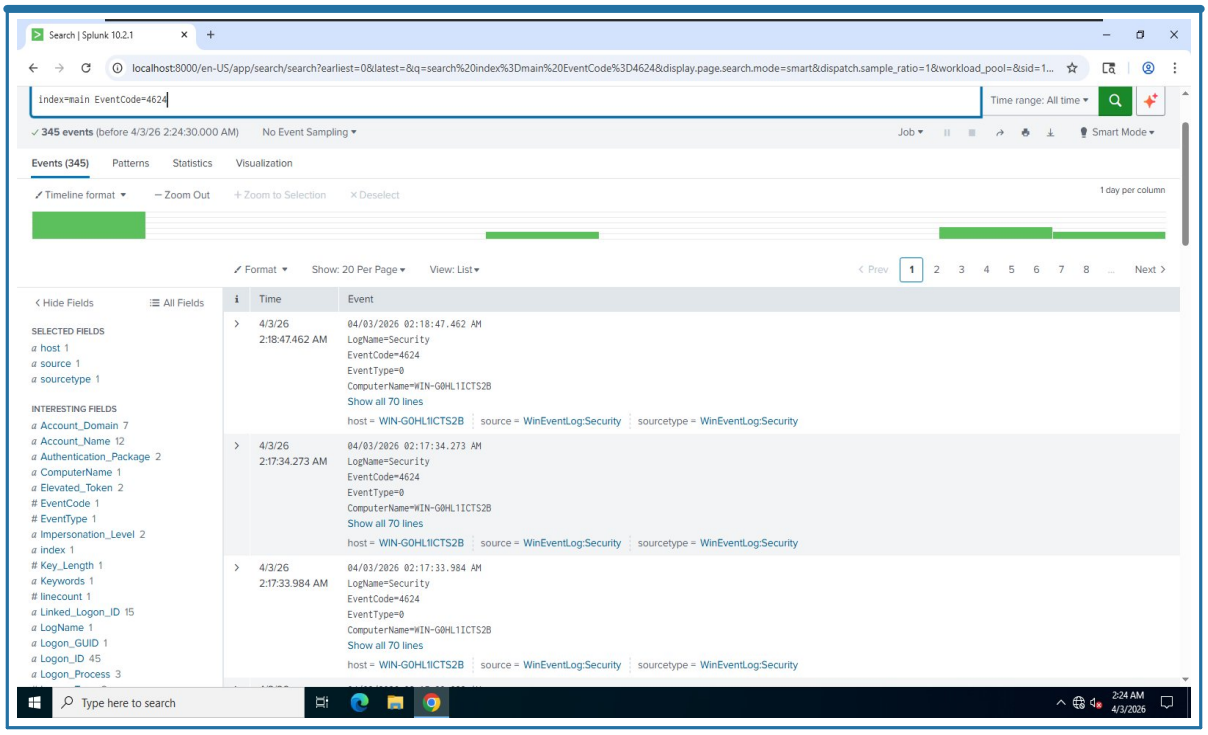

Figure 2 — EventCode=4624 query returning 345 successful logon events from WinEventLog:Security, establishing the authentication baseline.

SECTION 06

Step 3 — Brute Force Attack Simulation

MITRE ATT&CK | T1110 — Brute Force	MITRE ATT&CK | T1078 — Valid Accounts

A credential brute force attack was launched from the Kali Linux attacker machine using Hydra, targeting the RDP service (port 3389) on Windows Server 2022. The attack cycled through the *rockyou.txt* wordlist containing over 14 million password combinations against the administrator account — simulating the exact attack pattern a SOC analyst would encounter in a real enterprise environment.

ATTACK COMMAND EXECUTED:

```
hydra -l administrator -P /usr/share/wordlists/rockyou.txt 192.168.52.139 rdp -V -t 4
```

Parameter	Value	Description
`-l`	`administrator`	Target username
`-P`	`/usr/share/wordlists/rockyou.txt`	Password wordlist (14M entries)
`IP`	`192.168.52.139`	Target — Windows Server 2022
`rdp`	`Port 3389`	Protocol — Remote Desktop
`-t`	`4`	Parallel threads

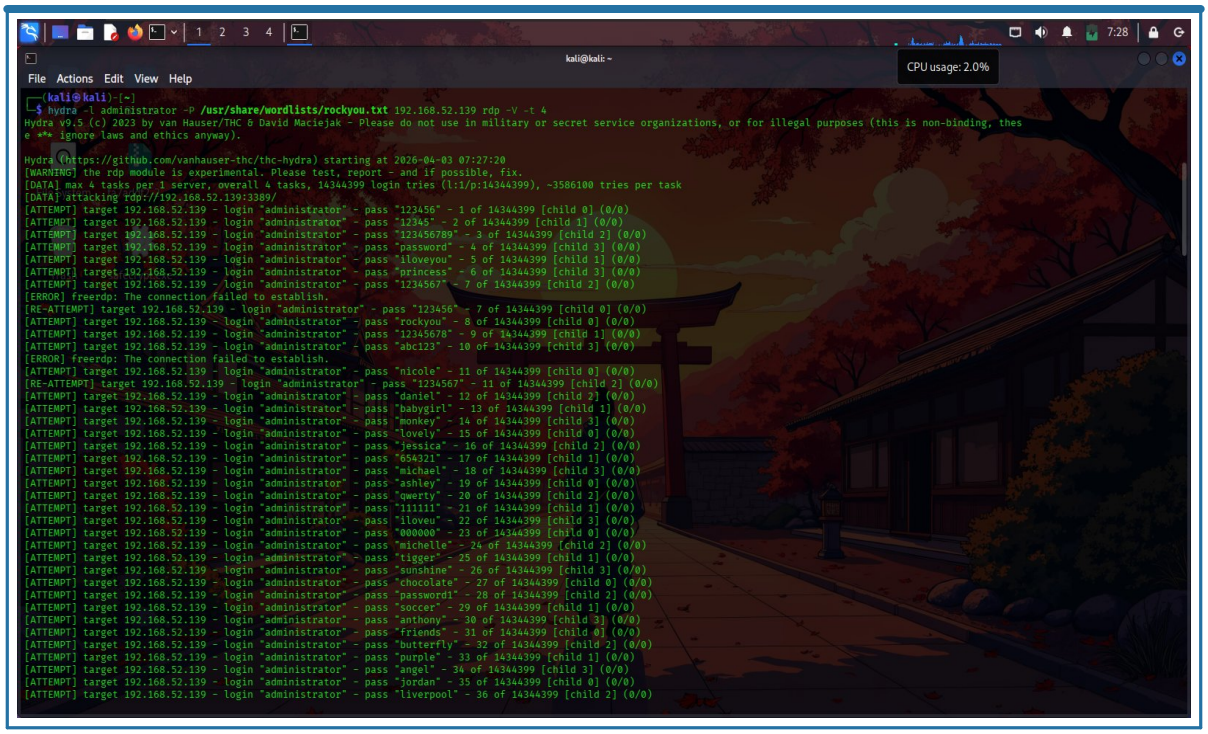

Figure 3 — Hydra brute force attack active on Kali Linux, making hundreds of RDP login attempts per minute against 192.168.52.139.

SECTION 07

Step 4 — Attack Detection (EventCode 4625)

Following the attack, Splunk was queried for **EventCode 4625** (Failed Logon). The search returned **626 failed authentication events** — all concentrated within an 8-minute window between **04:27 and 04:33 AM**. In a real SOC environment, this volume and velocity of failed logins from a single source would immediately trigger a high-priority alert and escalation.

DETECTION QUERY:

```
index=main source="WinEventLog:Security" EventCode=4625
```

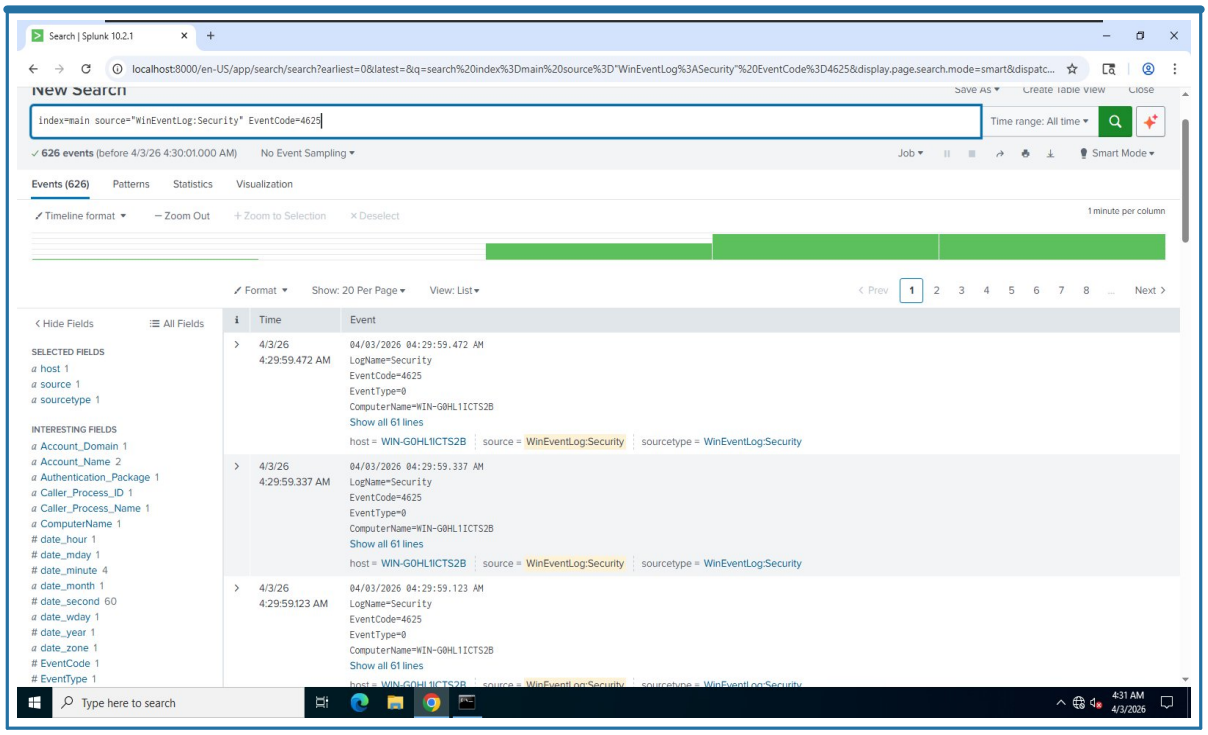

Figure 4 — Splunk returning 626 EventCode=4625 failed logon events from the Security log, all generated during the Hydra attack window.

SECTION 08

Step 5 — Attacker Identification via Log Correlation

Log correlation was used to identify the source of all failed login attempts. By grouping EventCode 4625 events by `Workstation_Name`, the investigation conclusively determined that all **3,326 failed attempts** originated from the workstation named **'kali'** — identifying the Kali Linux attacker machine beyond any doubt. This demonstrates the core SOC analyst capability of tracing an attack back to its origin through structured log correlation.

CORRELATION QUERY:

```
index=main source="WinEventLog:Security" EventCode=4625 | stats count by Workstation_Name |
sort -count
```

Workstation Name	Failed Login Count	Assessment
kali	3,326	**CONFIRMED ATTACKER — Kali Linux machine**

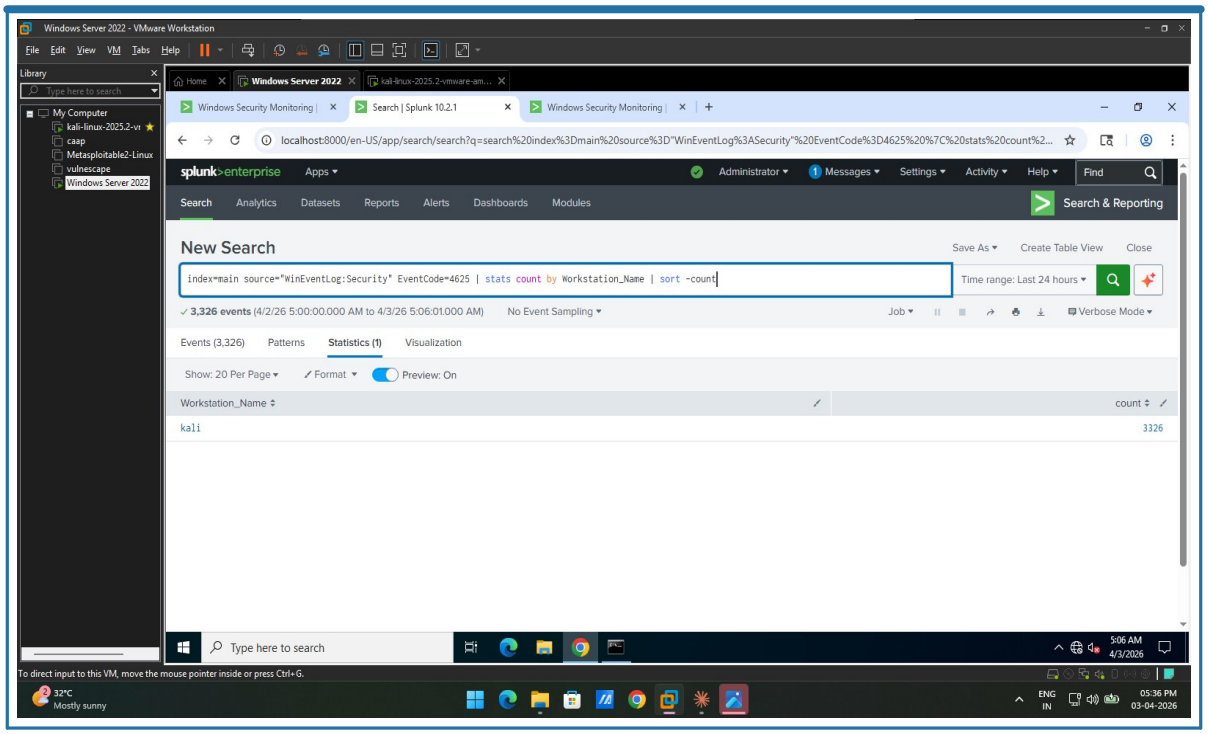

Figure 5 — Splunk log correlation confirms workstation 'kali' as the source of all 3,326 failed login attempts.

SECTION 09

Step 6 — Process Creation Analysis (EventCode 4688)

To determine whether the attacker executed any malicious processes during the attack window, all **EventCode 4688** (Process Creation) events were investigated. **44 events** were identified — every process was a legitimate Windows System32 executable, confirming no malicious code execution occurred and the attack was limited to credential brute force only.

PROCESS CREATION QUERY:

```
index=main source="WinEventLog:Security" EventCode=4688 | stats count by New_Process_Name |
sort -count | head 10
```

Process Name	Count	Assessment
C:\Windows\System32\smss.exe	12	**Legitimate — Session Manager**
C:\Windows\System32\csrss.exe	8	**Legitimate — Client Server Runtime**
C:\Windows\System32\autochk.exe	4	**Legitimate — Disk Check**
C:\Windows\System32\lsass.exe	4	**Legitimate — Local Security Authority**
C:\Windows\System32\services.exe	4	**Legitimate — Service Control Manager**
C:\Windows\System32\winlogon.exe	4	**Legitimate — Windows Logon**

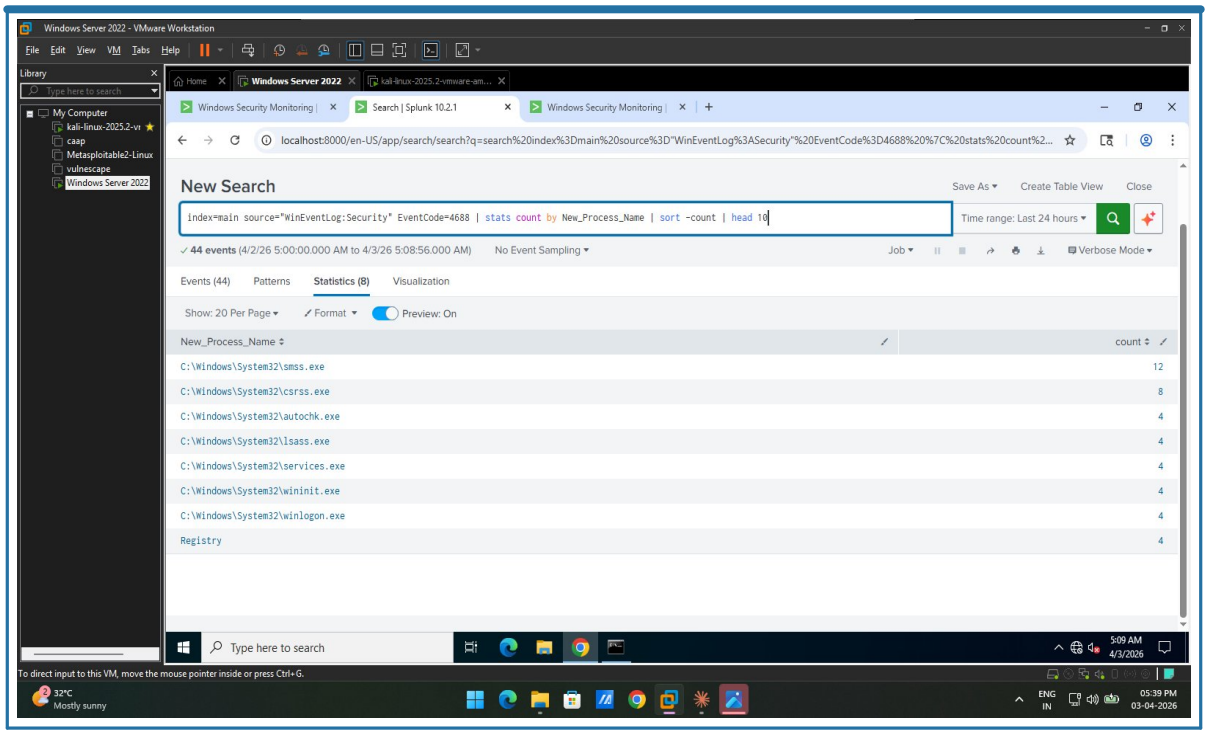

Figure 6 — EventCode=4688 analysis confirming all 44 process creation events are legitimate Windows System32 executables — no malicious execution detected.

SECTION 10

Step 7 — SOC Monitoring Dashboard

A professional four-panel 'Windows Security Monitoring' dashboard was built in Splunk to provide continuous real-time visibility into authentication and process activity. The dashboard is designed to reflect how a real enterprise SOC monitors multiple event types simultaneously, enabling rapid anomaly detection without manual searches.

Panel	Title	Event Source	Purpose
1	Successful Login Events	EventCode 4624	Monitor normal authentication baseline
2	Failed Login Events (Spike)	EventCode 4625	Detect brute force — spike clearly visible
3	Process Creation Events	EventCode 4688	Monitor for suspicious process execution
4	Top Targeted Accounts	EventCode 4625	Identify which accounts are under attack

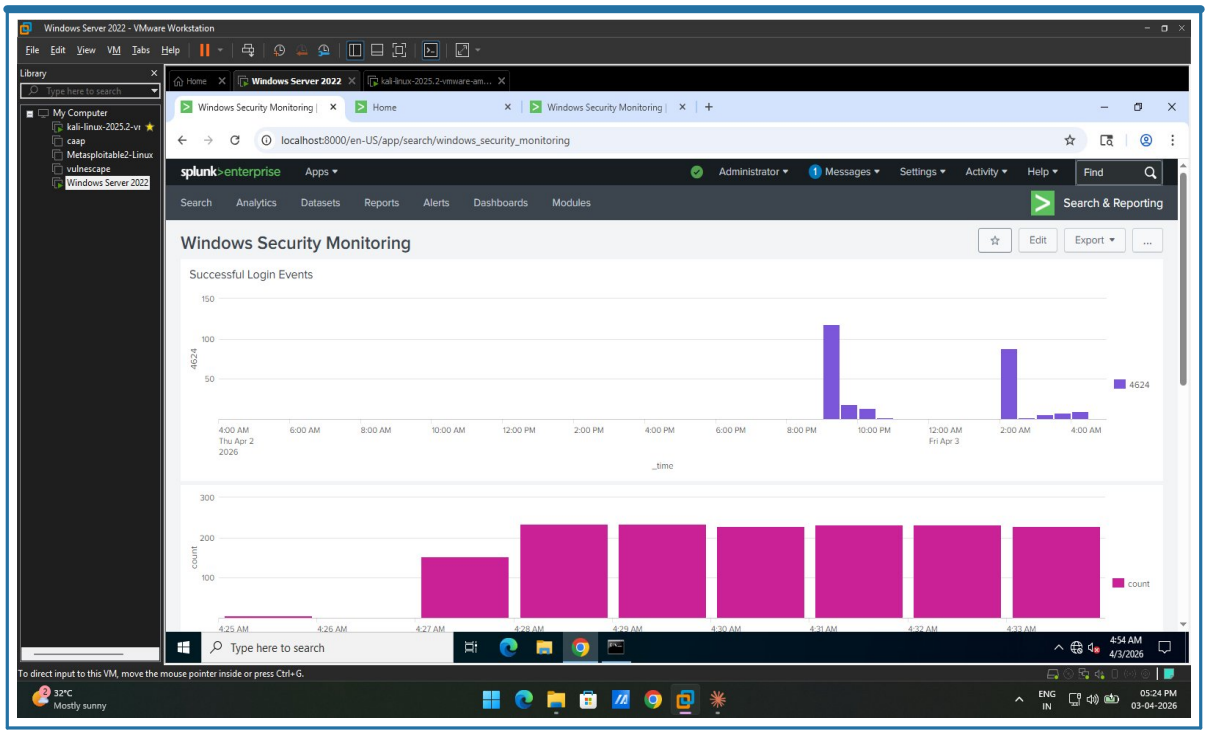

Figure 7 — Dashboard top panels: Successful login activity (purple) and the brute force attack spike (pink) clearly visible at 04:27–04:33 AM.

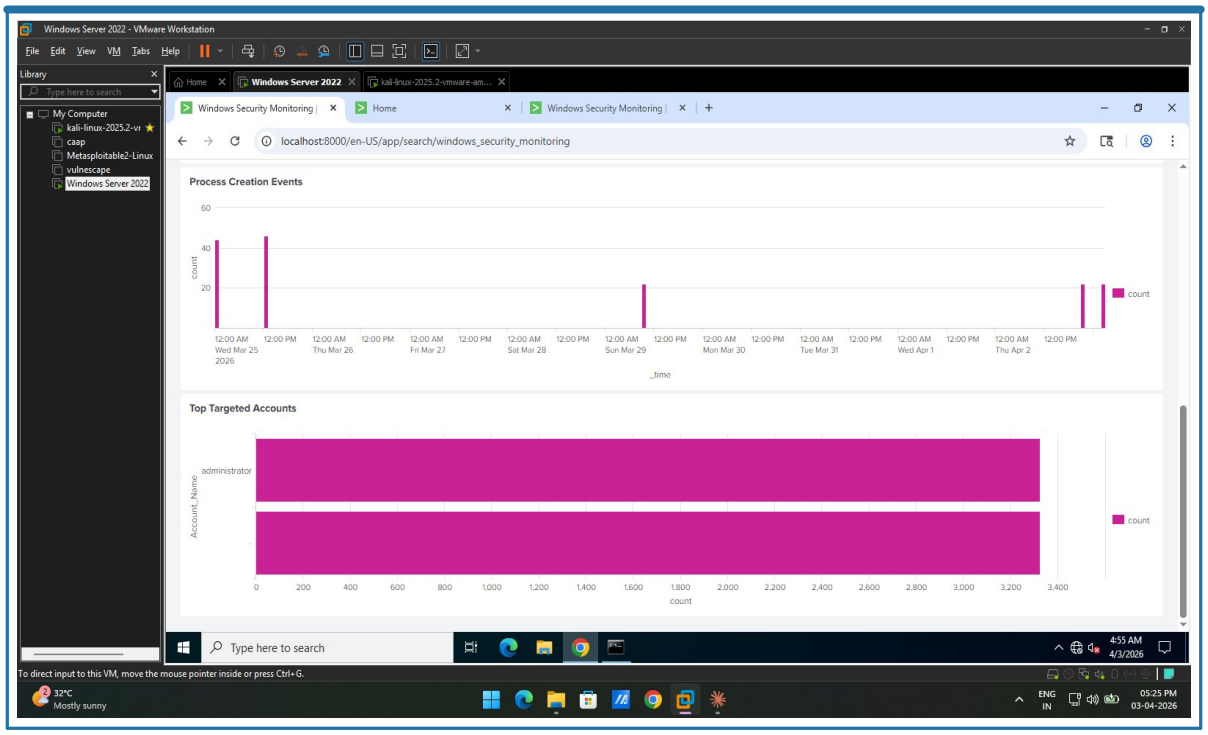

Figure 8 — Dashboard bottom panels: Process Creation Events and Top Targeted Accounts — administrator account hit 3,200+ times during the attack.

SECTION 11

Attack Detection Summary

Attack Type	Tool	Events Generated	MITRE ATT&CK	Outcome
RDP Brute Force	Hydra 9.5	626 × EventCode 4625	T1110 — Brute Force	DETECTED
Credential Attack	Hydra 9.5	3,326 failed attempts	T1078 — Valid Accounts	BLOCKED

POST-ATTACK VERIFICATION

Successful Login Analysis (EventCode 4624)

After identifying the attack, a post-incident analysis confirmed no accounts were compromised. All 267 successful logins belonged to legitimate system accounts — confirming the brute force attack was unsuccessful and no unauthorized access occurred.

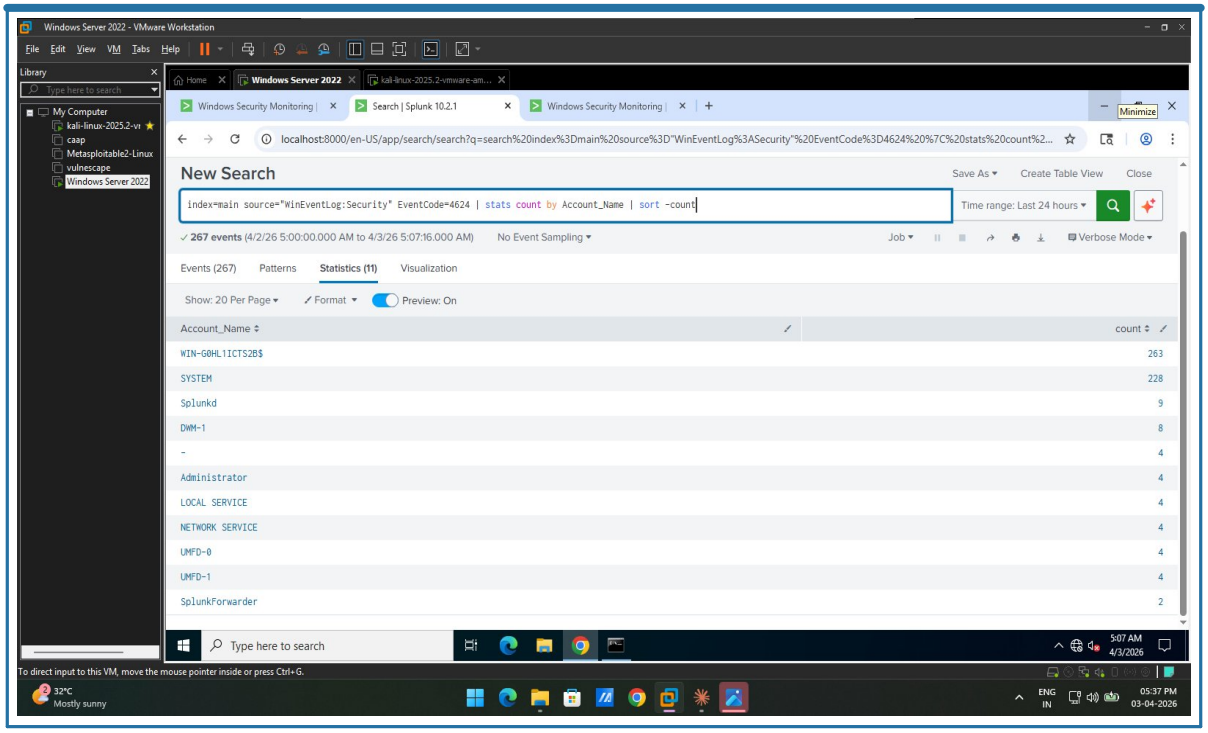

Figure 9 — Post-attack analysis: 267 successful logins across legitimate system accounts only — no unauthorized access confirmed.

SECTION 12

Splunk Queries Reference

Query 01 — Verify Security log ingestion

```
index=main source="WinEventLog:Security"
```

Query 02 — Successful login baseline (EventCode 4624)

```
index=main source="WinEventLog:Security" EventCode=4624
```

Query 03 — Detect failed login attempts (EventCode 4625)

```
index=main source="WinEventLog:Security" EventCode=4625
```

Query 04 — Monitor process creation (EventCode 4688)

```
index=main source="WinEventLog:Security" EventCode=4688
```

Query 05 — Identify attack source by workstation

```
index=main source="WinEventLog:Security" EventCode=4625 | stats count by Workstation_Name |
sort -count
```

Query 06 — Visualize brute force timeline

```
index=main source="WinEventLog:Security" EventCode=4625 | timechart count span=1m
```

Query 07 — Account breakdown for successful logins

```
index=main source="WinEventLog:Security" EventCode=4624 | stats count by Account_Name |
sort -count
```

Query 08 — Top 10 processes created

```
index=main source="WinEventLog:Security" EventCode=4688 | stats count by New_Process_Name |
sort -count | head 10
```

SECTION 13

Tools Used

Tool	Version	Purpose
Windows Server 2022	Build 20348	Target machine — Windows Security Event Log generation
Splunk Enterprise	10.2.1	SIEM — log ingestion, SPL search, and dashboard
Kali Linux	2025.2	Attacker machine — brute force simulation
Hydra	9.5	Credential brute force attack tool (RDP)
VMware Workstation Pro	17	Virtualization — running all lab VMs

SECTION 14

Key Findings & Conclusion

F-01	3,467 Windows Security Event logs successfully ingested into Splunk SIEM from Windows Server 2022.
F-02	345 successful logon events (EventCode 4624) analyzed to establish an authentication baseline.
F-03	626 failed logon events (EventCode 4625) detected — all generated within an 8-minute attack window.
F-04	Log correlation traced all 3,326 failed attempts to attacker workstation "kali" (Kali Linux).
F-05	Administrator account was the sole target — hit over 3,200 times during the attack.
F-06	No successful unauthorized access confirmed — brute force attack was unsuccessful.
F-07	44 process creation events (EventCode 4688) analyzed — zero malicious processes detected.
F-08	Four-panel SOC monitoring dashboard built and validated in Splunk Enterprise.

This project successfully demonstrates a complete SOC analyst workflow applied to Windows Event Log analysis — from SIEM configuration and baseline establishment, through attack simulation, detection, and log correlation, to dashboard creation and incident closure. Every finding is backed by real Splunk data and screenshots, validating all claims made in the portfolio.